

Manfred Fong

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Professional Experience

Software Engineer, Contract – UAlbany Center for Technology in Government, Remote **Apr 2026 – Jun 2026**

- Built a full-stack remote configuration system for AI-enabled cameras and sensors deployed across a fleet of 6 MTA buses, eliminating the need for on-site device access during configuration updates.
- Developed 10+ Django REST Framework GET and PATCH endpoints enabling precise field-level remote configuration of IoT sensor devices across 9 global settings and 5 per-sensor parameters, ensuring a failed or partial update never corrupts a live device's active config.
- Built a Next.js operator dashboard replacing a manual configuration workflow for deployed IoT devices across an active transit network.

Application Developer, Contract – NextHome Platinum Properties, Port Washington, NY **Jan 2026 – Mar 2026**

- Built a Chrome extension automating lead status updates across 500+ weekly referrals for a team of 8 real estate agents, eliminating 40+ hours of manual data entry per week and freeing agents to focus on client outreach and closing deals.
- Reverse-engineered a closed React application's internal event system to enable reliable programmatic form submission. A solution no public API or documentation supported.
- Engineered a polling-based DOM traversal system handling virtual scroll pagination and async content updates across a high-volume referral pipeline, replacing brittle fixed-delay approaches that caused frequent failures.

Risk Analyst – The Merchant Marketplace, Lynbrook, NY

Sep 2024 – Nov 2025

- Underwrote 50+ merchant cash advance applications per day, leveraging SQL and proprietary CRM software to retrieve and analyze merchant financial data, bank activity, and credit information to support funding decisions.
- Used SQL to extract and analyze underwriting and portfolio data, identifying fraud indicators and high-risk applications while helping maintain 95%+ underwriting accuracy.

Key Projects and Leadership

ThrottleIQNYC – React, TypeScript, FastAPI, PostgreSQL, Vercel, Railway

github.com/BasicallyManny/ThrottleIQ | throttle-iq.vercel.app

- Analyzed over 240,000 NYC traffic collision records to identify high-risk motorcycle crash patterns, uncovering trends throughout the day, seasonality, and other key risk factors.
- Developed an interactive crash-density heatmap using MapLibre GL JS and deck.gl to visualize 10,000+ geolocated collision records, featuring viewport-constrained navigation and precise longitude/latitude-based incident mapping.
- Built an automated ETL pipeline handling data ingestion, cleaning, and normalization into a relational schema, with monthly scheduled reloads via Railway cron jobs processing 20000+ records per run.

Institute of Electrical and Electronics Engineers (IEEE) – UAlbany Student Branch, Albany, NY **Oct 2022 – Aug 2024**

- Co-founded Hackadamian, UAlbany's first annual hackathon, coordinating logistics and technical infrastructure.
- Designed and built a public-facing event registration platform in React and Tailwind CSS supporting participant sign-ups and real-time event-day check-in.
- Delivered live technical support and troubleshooting for non-technical attendees during events and workshops, sharpening client-facing communication skills through direct support of 50+ students.

Certifications

- AWS Certified Cloud Practitioner (CLF-C02), Amazon Web Services — August 2026

Education

Bachelor of Science in Computer Science, Minor in Informatics – University at Albany, NY

2020 – 2024

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, HTML/CSS

Frameworks and Libraries: React, Next.js, FastAPI, Django REST Framework, Express.js, Tailwind CSS, Docker, Bootstrap

Tools and Databases: MySQL, PostgreSQL, MongoDB, Google Cloud, Railway, Vercel, Git, GitHub

Additional Skills: REST API Design, Data Validation, Financial Analysis, Underwriting